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Detecting and estimating changes in dependent functional data

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ABSTRACT

Change point detection in sequences of functional data is examined where the functional observations are dependent. Of particular interest is the case where the change point is an epidemic change (a change occurs and then the observations return to baseline at a later time). The theoretical properties for various tests for at most one change and epidemic changes are derived with a special focus on power analysis. Estimators of the change point location are derived from the test statistics and theoretical properties are investigated.

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1. Introduction

The statistical analysis of functional data has progressed rapidly over the past few years, leading to the possibility of more complex structures being amenable to such techniques. This is particularly true of the complex correlation structure present within and across many functional observed data, requiring methods that can deal both with internal and external dependences between the observations. Nonparametric techniques for the analysis of functional data are becoming well established (see Ferraty and Vieu [11] or Horváth and Kokoszka [15] for a good overview), and this paper sets out a nonparametric framework for change point analysis within dependent functional data. This extends the work of Berkes et al. [7] and Aue et al. [4] in the i.i.d case as well as of Hörmann and Kokoszka [14] for weakly dependent data, all of them for at most one change point (AMOC). In the present paper, a wide class of dependency structures is accounted for and two types of change point alternatives are considered, AMOC and epidemic changes, where the observations having changed return to their original state after some unknown time.

Tests and estimators are usually based on dimension-reduction techniques, where it is important that the change is not orthogonal to the projection subspace (for details see Sections 3.2 and 3.3). Most of the methodologies, including those in the references given above, choose this subspace based on estimated principal components. While the theory for change point tests developed in Section 3 is not limited to dimension-reduction techniques based on principal components, we will show in Section 4 why using principal components will lead to an improved power behavior of the test statistics. In fact a large enough change will switch the estimated principal components in such a way that the change is no longer orthogonal to the

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projection subspace making it detectable (cf. Theorem 4.1). This switch occurs even for small changes if the underlying covariance structure of the functional data is flat showing that this method yields good results even and especially for underlying covariance structures that are usually seen as being inappropriate for standard principal component analysis. In addition, we characterize detectable changes in terms of the (unobserved) uncontaminated covariance structure, formalizing remarks given in Berkes et al. [7].

The paper proceeds as follows. First, we introduce the change point problem in Section 2 in addition to summarizing some facts about functional time series and principal component analyses, which will be the key to the asymptotic properties of our change point detection procedures. In Section 3, methods for the detection and estimation of change points for dependent functional observations are derived. These methods are presented using an arbitrary orthonormal projection subspace which allows the same general theory to apply regardless of the subspace projection choice. In Section 4 we turn once again to principal component analysis and show why this is a good choice for dimension reduction for mean change detection. The final section gives the details of the proofs.

2. Preliminaries on functional data and principal component analysis

In this section, we will introduce the problem and summarize some known results on functional time series and principal components which are needed to obtain asymptotic properties of the change point procedures.

2.1. Change point problem

We consider a mean change problem in a series of functional observations $X_i(t)$, $t \in \mathcal{Z}$, $i = 1, \ldots, n$, where $\mathcal{Z}$ is some compact set. While in statistics almost any data set is being taken discretely for numerical reasons, the true underlying observations can often be assumed to be a (smooth) functional observation on a compact set. Examples include brain data (confer Aston and Kirch [3]), temperature data (confer Berkes et al. [7]) or high-frequency financial data (confer Aue et al. [5]). Nevertheless, even in non-functional but very high-dimensional settings standard multivariate change point procedures may not be numerically stable due to the necessity of accurately estimating the inverse of the covariance or even long-run covariance matrix, so that the dimension-reduction techniques developed in this paper will also be useful in this setting. In fact, all of our theoretic results remain true in a multivariate setting which is somewhat easier to treat due to the finite-dimensional basis.

The simplest mean change model for functional data is given by the at most one change (AMOC) model

$$X_i(t) = Y_i(t) + \mu_1(t)1_{\{i \leqslant \vartheta n\}} + \mu_2(t)1_{\{\vartheta n < i \leqslant n\}}, \tag{2.1}$$

where the mean functions before and after the change $\boldsymbol{\mu}_j = \mu_j(\cdot)$ as well as the functional time series $\{Y_i(\cdot) : 1 \leqslant i \leqslant n\}$ are elements of $L^2(\mathcal{Z})$, $0 < \vartheta \leqslant 1$ describes the position of the change, $\mathrm{E}Y_i(t) = 0$. $\boldsymbol{\mu}_1$, $\boldsymbol{\mu}_2$ as well as ϑ are unknown.

Throughout the paper we will assume:

Assumption $\mathcal{P}$.1. The time series $\{Y_i(\cdot) : i \geqslant 1\}$ is centered, stationary and ergodic with

$$\mathrm{E}\|Y_1(\cdot)\|^2 = \int \mathrm{E}(Y_1^2(t))\, dt < \infty.$$

This setting for independent (functional) observations with at most one change point (AMOC) was investigated by Berkes et al. [7] and for specific weak dependent processes by Hörmann and Kokoszka [14]. We will also allow for dependency (in time) of the functional observations pointing out what properties on the time series are needed to obtain the desired asymptotic results. This allows the reader to generalize the results to any weak dependency concept fulfilling those properties.

In this paper we consider a more complicated change point model, namely an epidemic change model, where after a certain time the mean changes back. In many applications (such as regulation of gene expression) this is the type of change that can be expected. The epidemic model is given by

$$X_i(t) = Y_i(t) + \mu_1(t) + (\mu_2(t) - \mu_1(t))1_{\{\vartheta_1 n < i \leqslant \vartheta_2 n\}}, \tag{2.2}$$

where $\boldsymbol{\mu}_j$ and $\{Y_i(\cdot) : 1 \leqslant i \leqslant n\}$ are as above, $0 < \vartheta_1 \leqslant 1$ marks the beginning of the epidemic change, while $\vartheta_1 \leqslant \vartheta_2 \leqslant 1$ marks the end of the epidemic change. $\boldsymbol{\mu}_1$, $\boldsymbol{\mu}_2$ as well as ϑ_1, ϑ_2 are unknown.

2.2. Some properties of functional time series

In this section we will summarize some results obtained for certain weak dependent functional time series that will be the key to the asymptotic proofs in Section 3. While these key properties will certainly hold for many more weak dependent time series, we restrict the discussion in this paper to strong mixing time series as well as $L^p - m$-approximable functional time series as introduced by Hörmann and Kokoszka [14]. Let us recall the definition of these dependency concepts.

Definition 2.1. (a) A stationary (possibly Hilbert-space valued) process $\{Y_j : j \in \mathbb{Z}\}$ is called strong mixing with mixing rate r_m if

$$\sup_{A,B} |P(A \cap B) - P(A)P(B)| = O(r_m), \qquad r_m \to 0,$$

where the supremum is taken over all $A \in \mathcal{A}(Y_0, Y_{-1}, \ldots)$ and $B \in \mathcal{A}(Y_m, Y_{m+1}, \ldots)$.

(b) A stationary (Hilbert-space valued) process $\{Y_j : j \in \mathbb{Z}\} \in L^p_H$ is called $L^p - m$-approximable if $Y_j = f(\epsilon_j, \epsilon_{j-1}, \ldots)$, where ϵ_i are i.i.d., f is measurable and

$$\sum_{m \geqslant 1} \left(\mathrm{E}\|Y_m - Y_m^{(m)}\|^p\right)^{1/p} < \infty,$$

where $Y_j^{(m)} = f(\epsilon_j, \ldots, \epsilon_{j-m+1}, \epsilon'_{j-m}, \epsilon'_{j-m+1}, \ldots)$ and $\{\epsilon'_j\}$ is an independent copy of $\{\epsilon_j\}$.

Strong mixing conditions yield very sharp results and have been widely used in statistics (for a complete account of the classic theory we refer to Bradley [9]). However, they are often hard to verify in practice and exclude some examples that are important in statistics such as certain AR(1)-time series with discrete innovations (confer Andrews [1]). For these reasons many new weak dependency concepts have been developed in recent years and it is too early to tell which one will play the dominant role in the future. This motivates us to give certain basic results for time series of the above two types in this section but then to develop the change-point theory in Section 3 based merely on those results enabling future researchers to apply them to different weak dependency concepts as long as the same basic results hold.

Most statistical methodology for functional or very high-dimensional data relies on projections into a lower-dimensional space of dimension d such as

$$\eta_{i,l} = \int Y_i(t) v_l(t)\, dt \quad i = 1, \ldots, n,\ l = 1, 2, \ldots, d, \tag{2.3}$$

where $v_1(\cdot), \ldots, v_d(\cdot)$ is an orthonormal system with respect to the $L^2(\mathcal{Z})$-norm.

Lemma 2.1. *Let $\{Y_i(\cdot)\}$ be either $L^2 - m$-approximable or strong mixing with $\mathrm{E}\|Y_i(\cdot)\|^{2+\delta} < \infty$ for some $0 < \delta \leqslant 1$ and mixing rate $r_m = m^{-c}$, $c > (2+\delta)/\delta$. Then the following assertions hold for $\boldsymbol{\eta}_i = (\eta_{i,1}, \ldots, \eta_{i,d})^T$ with $\eta_{i,l}$ as in (2.3), $l = 1, \ldots, d$:*

(a) *The time series $\{\boldsymbol{\eta}_i : i \in \mathbb{Z}\}$ is stationary and short-range dependent i.e.*

$$\sum_{i \in \mathbb{Z}} |\mathrm{cov}(\eta_{0,l_1}, \eta_{i,l_2})| < \infty, \quad l_1, l_2 = 1, \ldots, d. \tag{2.4}$$

(b) *$\{\boldsymbol{\eta}_i\}$ fulfills the following functional limit theorem*

$$\left\{\frac{1}{\sqrt{n}} \sum_{1 \leqslant i \leqslant nx} \boldsymbol{\eta}_i : 0 \leqslant x \leqslant 1\right\} \xrightarrow{D^d[0,1]} \{\Sigma^{1/2} \boldsymbol{W}_d(x) : 0 \leqslant x \leqslant 1\}, \tag{2.5}$$

where $\boldsymbol{W}_d$ is a d-dimensional process whose components are independent Wiener processes and $\Sigma = \sum_{k \in \mathbb{Z}} \Gamma(k)$ is a positive definite matrix with $\Gamma(h) = \mathrm{E}\boldsymbol{\eta}_t \boldsymbol{\eta}_{t+h}^T$, $h \geqslant 0$, and $\Gamma(h) = \Gamma(-h)^T$ for $h < 0$.

Finally, we show that the projections of mixing resp. $L^4 - m$-approximable sequences fulfill a Hájek-Rényi-type inequality.

Lemma 2.2. *Let $\{\xi_i(\cdot)\}$ be a centered real time series that is either $L^4 - m$-approximable with*

$$\sup_{k,l \geqslant 0} \sum_{r=1}^{\infty} \left|\mathrm{cov}\left(\xi_0(\xi_k - \xi_k^{(k)}), \xi_r^{(r)} \xi_{r+l}^{(r+l)}\right)\right| < \infty \tag{2.6}$$

or strong mixing with $\mathrm{E}\|\xi_i(\cdot)\|^{2+\delta} < \infty$ for some $0 < \delta \leqslant 1$ and mixing rate $r_m = m^{-c}$, $c > (2+\delta)/\delta$. Then, there exists an increasing sequence $\alpha(n) \to \infty$ such that ξ_i fulfill the following Hájek-Rényi-type inequalities:

$$\max_{1 \leqslant k \leqslant n} \frac{\alpha(k)}{k} \left|\sum_{i=1}^{k} \xi_i\right| = O_P(1), \quad l = 1, \ldots, d,$$

$$\max_{1 \leqslant k \leqslant n} \frac{\alpha(n-k)}{n-k} \left|\sum_{i=k+1}^{n} \xi_i\right| = O_P(1), \quad l = 1, \ldots, d. \tag{2.7}$$

In Section 3.3 we will use the above Hájek-Rényi inequalities for $\xi_i = \int Y_i(t) v(t)\, dt$ for some L^2-function $v(t)$ and $Y_i(\cdot)$ as above. To this end note that strong mixing ($L^4 - m$-approximability) of $\{Y_i(\cdot)\}$ implies that $\{\xi_i\}$ is strong-mixing ($L^4 - m$-approximable) as can be seen by the proof of Lemma 2.1. Furthermore ξ_i is centered if $Y_i(\cdot)$ is centered. Condition (2.6) is a technical condition related to classic cumulant summability conditions. Hörmann and Kokoszka [14] show that it always holds for linear processes and give some motivation why it is not a strong condition even for nonlinear sequences.

2.3. Projections using principal components

Classical dimension reduction techniques are often based on the first d principal components, which define a subspace of dimension d explaining the most variation of any subspace of size d. We will shortly describe the main ideas in this section as well as some properties of principal components needed for the asymptotics of the change point procedures in Section 3. In Section 4 we will discuss why principal components are especially suitable as a dimension reduction technique in the context of mean change analysis.

Define the covariance kernel of $Y_i(\cdot)$ given by

$$c(t,s) = \mathrm{E}(Y_i(t)Y_i(s)). \tag{2.8}$$

The covariance operator $C : \mathcal{L}^2(\mathcal{Z}) \to \mathcal{L}^2(\mathcal{Z})$ is obtained as $Cz = \int_{\mathcal{Z}} c(\cdot,s)z(s)\,ds$. Due to the stationarity of $\{Y_i(\cdot) : 1 \leqslant i \leqslant n\}$ the covariance kernel does not depend on i and is square integrable due to the Cauchy–Schwarz inequality as well as the square integrability of $Y_1(\cdot)$.

Let $\{\lambda_k\}$ be the non-negative decreasing sequence of eigenvalues and $\{v_k(\cdot) : k \geqslant 1\}$ a given set of corresponding orthonormal eigenfunctions of the covariance operator, i.e. they are defined by

$$\int c(t,s)v_l(s)\,ds = \lambda_l v_l(t), \quad l = 1,2,\ldots,\ t \in \mathcal{Z}. \tag{2.9}$$

Under the above assumptions, the covariance kernel can be written as

$$c(t,s) = \sum_{k=1}^{\infty} \lambda_k v_k(t) v_k(s),$$

and more importantly $Y_i(\cdot)$ can be expressed in terms of the eigenfunctions

$$Y_i(t) = \sum_{l=1}^{\infty} \eta_{i,l} v_l(t), \tag{2.10}$$

where $\{\eta_{i,l} : l = 1,2,\ldots\}$ are uncorrelated with mean 0 and variance λ_l for each i.

The infinite sum on the right-hand side converges in $L^2(\mathcal{Z})$ with probability one.

The fact that the scores are uncorrelated is useful for the change point analysis below in case of independent data as we can more easily estimate Σ due to its diagonal structure in the independent case. Unfortunately, for dependent functional data this is no longer true in general as the long-run covariance can be different from zero even if η_{i,l_1} and η_{i,l_2} are uncorrelated for any i.

The scores can be calculated as in (2.3) where the eigenfunctions form an orthonormal system.

More details on functional principal component analysis can be found in the papers by Hall and Hosseini-Nasab [13] and Benko et al. [6] or the books by Bosq [8] and Horváth and Kokoszka [15].

In practice, the covariance kernel $c(t,s)$ is usually not known but needs to be estimated. A natural estimator in a general non-parametric setting is the empirical version of the covariance function

$$\widehat{c}_n(t,s) = \frac{1}{n}\sum_{i=1}^{n}(X_i(t) - \bar{X}_n(t))(X_i(s) - \bar{X}_n(s)), \tag{2.11}$$

where $\bar{X}_n(t) = \frac{1}{n}\sum_{i=1}^{n} X_i(t)$.

The following lemma shows that this estimator is consistent with a certain rate if the sequence $\{X_i(\cdot)\}$ is stationary, which corresponds to the null hypothesis $X_i(\cdot) = \mu_1(\cdot) + Y_i(\cdot)$ in the change point model.

Lemma 2.3. *Let $X_i(\cdot) = \mu_1(\cdot) + Y_i(\cdot)$ be a stationary sequence with covariance kernel $c(t,s)$.*

(a) *If $\{Y_i(\cdot) : i \geqslant 1\}$ fulfills Assumption $\mathcal{P}$.1, then*

$$\iint (\widehat{c}_n(t,s) - c(t,s))^2\,dt\,ds = o_P(1).$$

(b) *If $Y_i(\cdot)$ is additionally $L^4 - m$-approximable or strong mixing with mixing rate r_j, $\sum_{h\geqslant 1} r_h^{\frac{\delta}{4+\delta}} < \infty$ and $\mathrm{E}\|Y_1(\cdot)\|^{4+\delta} < \infty$, then*

$$\iint (\widehat{c}_n(t,s) - c(t,s))^2\,dt\,ds = O_P(n^{-1}). \tag{2.12}$$

When we apply this estimator in the change point situation with a mean change present we can no longer expect that it converges to the covariance kernel $c(t,s)$ of $Y_i(\cdot)$, but it will converge to a different contaminated limit $k(t,s)$ as the following lemma shows.

Lemma 2.4. *If $\{Y_i(\cdot) : i \geqslant 1\}$ fulfills Assumption $\mathcal{P}$.1 and $X_i(\cdot)$ follows one of the mean change models (2.1) or (2.2), then*

$$\iint (\widehat{c}_n(t,s) - k(t,s))^2 \, dt \, ds \xrightarrow{P} 0, \tag{2.13}$$

where

$$k(t,s) = c(t,s) + \theta(1-\theta)\Delta(t)\Delta(s), \tag{2.14}$$

and

$$\Delta(t) = \mu_1(t) - \mu_2(t),$$
$$\theta = \begin{cases} \vartheta, & \text{AMOC}, \\ \vartheta_2 - \vartheta_1, & \text{epidemic change}. \end{cases}$$

From the above two lemmas we make conclusions on the convergence rate of the corresponding estimated eigenfunctions (eigenvalues) to the eigenfunctions (eigenvalues) of the respective limit kernels $c(t,s)$ or $k(t,s)$.

Theorem 2.1. *Let*

$$\iint (\widehat{c}_n(t,s) - \tilde{c}(t,s))^2 \, dt \, ds = o_P(1).$$

Denote by $\widehat{\lambda}_k$ and $\widehat{v}_k(\cdot)$ the eigenvalues (in decreasing order) and corresponding orthonormal eigenfunctions of $\widehat{c}_n(t,s)$ and by $\tilde{\lambda}_k$ and $\tilde{v}_k(\cdot)$ the eigenvalues (in decreasing order) and corresponding orthonormal eigenfunctions of $\tilde{c}(t,s)$. Additionally, we assume that $\tilde{\lambda}_1 > \tilde{\lambda}_2 > \cdots > \tilde{\lambda}_d > \tilde{\lambda}_{d+1}$.

(a) *Then, it holds for $j = 1, \ldots, d$*

$$|\widehat{\lambda}_j - \tilde{\lambda}_j| \xrightarrow{P} 0, \qquad \int (\widehat{v}_j(t) - \tilde{s}_j \tilde{v}_j(t))^2 \, dt \xrightarrow{P} 0,$$

where $\tilde{s}_j = \operatorname{sgn}(\int \tilde{v}_j(t)\widehat{v}_j(t)\, dt)$.

(b) *If additionally the following rate of convergence holds*

$$\iint (\widehat{c}_n(t,s) - \tilde{c}(t,s))^2 \, dt \, ds = O_P(n^{-1}),$$

then we get

$$|\widehat{\lambda}_j - \tilde{\lambda}_j| = O_P(n^{-1/2}), \qquad \int (\widehat{v}_j(t) - \tilde{s}_j \tilde{v}_j(t))^2 \, dt = O_P(n^{-1}).$$

The above theorem holds for any estimator $\widehat{c}_n(t,s)$ fulfilling the assumptions of the theorem and is not restricted to the example in (2.11). Furthermore, it also applies to the misspecified situation with the contaminated limit $\tilde{c}(t,s) = k(t,s)$.

The assumption on the eigenvalues is standard in principal component analysis and guarantees that the orthonormal eigenfunctions are identifiable up to their sign which is the reason why $\tilde{s}_j$ is required in the theorem.

3. Change point detection procedures

In this section we develop some general theory for change point detection procedures. We do not make any assumptions on the dependency present within the stationary sequence $Y_i(\cdot)$ but rather emphasize the critical properties which are needed to obtain the asymptotic results hence allowing easy extensions to different dependency concepts. However, the theory stated in Section 2.2 shows that all of the theory in this section holds for certain mixing sequences as well as $L^4 - m$-approximable sequences. In a similar spirit, we do not require the projection into a lower dimensional space to be based on principal components of the estimator (2.11) but allow for arbitrary projections. Again, the theory developed in Section 2.3 gives all necessary results for principal components using estimator (2.11) for strong mixing and $L^4 - m$-approximable sequences (which implies $L^2 - m$-approximability by an application of the Cauchy–Schwarz inequality).

3.1. Testing statistics and null asymptotics

First, we will consider the testing problem of the null hypothesis of no change in the mean

$$H_0 : \mathrm{E}X_i(\cdot) = \mu_1(\cdot), \quad i = 1, \ldots, n,$$

versus the AMOC alternative

$$H_1^{(A)}: \mathrm{E}X_1(\cdot) = \mu_1(\cdot), \quad i = 1, \ldots, \lfloor \vartheta n \rfloor, \quad \text{but}$$
$$\mathrm{E}X_1(\cdot) = \mu_2(\cdot) \neq \mu_1(\cdot), \quad i = \lfloor n\vartheta \rfloor + 1, \ldots, n, \ 0 < \vartheta < 1$$

respectively versus the epidemic change alternative

$$H_1^{(B)}: \mathrm{E}X_1(\cdot) = \mu_1(\cdot), \quad i = 1, \ldots, \lfloor \vartheta_1 n \rfloor, \lfloor \vartheta_2 n \rfloor + 1, \ldots, n, \quad \text{but}$$
$$\mathrm{E}X_1(\cdot) = \mu_2(\cdot) \neq \mu_1(\cdot), \quad i = \lfloor n\vartheta_1 \rfloor + 1, \ldots, \lfloor \vartheta_2 n \rfloor, \ 0 < \vartheta_1 < \vartheta_2 < 1.$$

Note that the null hypothesis corresponds to the cases where $\vartheta = 1$ (AMOC) resp. $\vartheta_1 = \vartheta_2 = 1$ (epidemic change).

It is well known how to test for AMOC mean changes in multivariate observations (cf. e.g. Horváth et al. [16]). However, in a functional setting, respectively for high-dimensional data, this is computationally infeasible. Here, the idea is to use a projection into a lower dimensional space and use standard change point statistics for the projected data. In Section 2.3 we already discussed the standard approach for dimension reduction based on estimated principal components, which was also used by Berkes et al. [7]. In Section 4 we discuss why this is a very favorable approach in the context of testing for changes in the mean with respect to the obtained power. Concerning tests for epidemic changes in a univariate as well as multivariate setting, we refer to Antoch and Hušková [2] respectively Jarušková and Piterbarg [18].

In the remainder of this Section 3.1 we assume that $\{v_k(\cdot), k = 1, \ldots, d\}$ and $\{\widehat{v}_k(\cdot), k = 1, \ldots, d\}$ are orthonormal systems. $\widehat{v}_k(\cdot)$, $k = 1, \ldots, d$, are estimators which fulfill under H_0

$$\int (\widehat{v}_k(t) - s_k v_k(t))^2 \, dt = O_P(n^{-1}), \quad s_k = \operatorname{sgn}\left(\int v_k(t)\, \widehat{v}_k(t)\, dt\right). \tag{3.15}$$

By Lemma 2.3 and Theorem 2.1 this holds in particular for estimated principal components used on strong mixing or $L^4 - m$-approximable time series.

We are now ready to explain the main idea of the testing procedure. Denote by $\widehat{\eta}_{i,l}$ the estimated scores, i.e. the projection coefficients of the estimated orthonormal system,

$$\widehat{\eta}_{i,l} = \langle X_i, \widehat{v}_l \rangle = \int X_i(t) \widehat{v}_l(t)\, dt, \quad i = 1, \ldots, n, \ l = 1, \ldots, d.$$

Then, $\widehat{\boldsymbol{\eta}}_i = (\widehat{\eta}_{i,1}, \ldots, \widehat{\eta}_{i,d})^T$ is a d-dimensional time series exhibiting an AMOC (epidemic) change if the functional sequence $\{X_i(\cdot) : 1 \leqslant i \leqslant n\}$ had an AMOC (epidemic) change as long as the change is not orthogonal to the subspace spanned by $\widehat{v}_1(\cdot), \ldots, \widehat{v}_d(\cdot)$. To see this, consider the stationary time series

$$\check{\eta}_{i,l} = \langle Y_i, \widehat{v}_l \rangle = \int Y_i(t) \widehat{v}_l(t)\, dt. \tag{3.16}$$

Then it holds

$$\widehat{\eta}_{i,l} = \check{\eta}_{i,l} + 1_{\{i \leqslant \vartheta n\}} \int \mu_1(t) \widehat{v}_l(t)\, dt + 1_{\{i > \vartheta n\}} \int \mu_2(t) \widehat{v}_l(t)\, dt \tag{3.17}$$

in case of AMOC change and an analogous expression for the epidemic change. Consequently, a change is present in the projected data if

$$\int \Delta(t) \widehat{v}_l(t)\, dt \neq 0, \quad \Delta(t) = \mu_1(t) - \mu_2(t), \ \text{for some } l = 1, \ldots, d.$$

This representation suggests to use multivariate change point statistics based on $\widehat{\boldsymbol{\eta}}_i$, $i = 1, \ldots, n$, which are usually based on

$$\frac{1}{\sqrt{n}} \mathbf{S}_n(k) = \frac{1}{\sqrt{n}} \left(\sum_{i=1}^{k} \widehat{\boldsymbol{\eta}}_i - \frac{k}{n} \sum_{i=1}^{n} \widehat{\boldsymbol{\eta}}_i \right). \tag{3.18}$$

If we use a complete orthonormal basis, we even obtain the Hilbert space analogue of the classic CUSUM change point statistic, since by $X_i = \sum_{l \geqslant 1} \widehat{\eta}_{i,l} \widehat{v}_l$ and Parseval's identity

$$\max_k \frac{1}{n} \sum_{l \geqslant 1} \left(\sum_{i=1}^{k} \left(\widehat{\eta}_{i,l} - \bar{\widehat{\eta}}_l \right) \right)^2 = \max_k \frac{1}{n} \left\| \sum_{i=1}^{k} (X_i(t) - \bar{X}_i(t)) \right\|^2, \tag{3.19}$$

where $\| \cdot \|$ is the L^2-norm.

The following lemma gives the null asymptotics in $D[0, 1]$ for the process $\mathbf{S}_n(\cdot)$. From this we can easily obtain the null asymptotics of various popular test statistics in our main Theorem 3.1.

Lemma 3.1. *Let the null hypothesis hold, i.e. $X_i(\cdot) = \mu_1(\cdot)+Y_i(\cdot)$ with $Y_1(\cdot)$ fulfilling $\mathcal{P}.1$ and let the estimators $\widehat{v}_k(\cdot)$ fulfill (3.15). Additionally, the projections $\eta_{i,l} = \int Y_i(t)v_l(t)\,dt,\ i = 1,\dots,n,\ l = 1,2,\dots,$ of $Y_i(\cdot)$ need to fulfill (2.4) and (2.5). Then (as $n \to \infty$)*

$$\left\{\frac{1}{\sqrt{n}}\widetilde{\mathbf{S}}_n(x) : 0 \leqslant x \leqslant 1\right\} \xrightarrow{D^d[0,1]} \left\{\Sigma^{1/2}\mathbf{B}_d(x) : 0 \leqslant x \leqslant 1\right\},$$

where $\widetilde{\mathbf{S}}_n(x) = (\widetilde{S}_{n,1}(x),\dots,\widetilde{S}_{n,d}(x))^T$, $s_l = \operatorname{sgn}(\int v_k(t)\widehat{v}_k(t)\,dt)$ and

$$\widetilde{S}_{n,l}(x) = s_l S_{n,l}(x) = s_l \sum_{1\leqslant i\leqslant nx}\left(\widehat{\eta}_{i,l} - \frac{1}{n}\sum_{i=1}^{n}\widehat{\eta}_{i,l}\right), \quad l = 1,\dots,d. \tag{3.20}$$

Σ is as in (2.4) and $\mathbf{B}_d$ is a d-dimensional Gaussian process whose components are independent Brownian bridges.

Remark 3.1. The proof shows that the result remains valid if the rate in (3.15) is replaced by $o_P(1)$ and we additionally assume that

$$\sup_{0<x<1}\int\left(\frac{1}{\sqrt{n}}\sum_{1\leqslant i\leqslant nx} Y_i(t)\right)^2 = O_P(1).$$

The latter one follows for example from functional central limit theorems for the Hilbert space valued random variables $\{Y_i(\cdot)\}$.

Since we do not know s_l, test statistics should be based on $\left(S_{n,l}(\cdot)\right)^2$ because $s_l^2 = 1$ by definition. The next theorem gives the null asymptotics for popular statistics. Analogous results for weighted versions of the statistics can also be obtained from Lemma 3.1. However, statistics with extreme-value type convergence do not fall into this framework but can be dealt with similarly as soon as strong invariance principles for the projections $\{\boldsymbol{\eta}_i : i = 1,\dots,n\}$ are available.

Theorem 3.1. *Let the assumptions of Lemma 3.1 hold. Furthermore let $\widehat{\Sigma}$ be a consistent symmetric positive-definite estimator for Σ (under H_0) and $B_l(\cdot)$, $l = 1,\dots,d$, be independent standard Brownian bridges.*

(a) *The following statistics are suitable to detect AMOC-change alternatives:*

$$T_n^{(A1)} = \frac{1}{n^2}\sum_{k=1}^{n}\mathbf{S}_n(k/n)^T\,\widehat{\Sigma}^{-1}\mathbf{S}_n(k/n),$$

$$T_n^{(A2)} = \max_{1\leqslant k\leqslant n}\frac{1}{n}\mathbf{S}_n(k/n)^T\,\widehat{\Sigma}^{-1}\mathbf{S}_n(k/n),$$

$$\text{where } \mathbf{S}_n(x) = \sum_{1\leqslant j\leqslant nx}\left(\widehat{\boldsymbol{\eta}}_j - \frac{1}{n}\sum_{i=1}^{n}\widehat{\boldsymbol{\eta}}_i\right).$$

Under H_0 it holds:

(i) $T_n^{(A1)} \xrightarrow{\mathcal{L}} \sum_{1\leqslant l\leqslant d}\int_0^1 B_l^2(x)\,dx$

(ii) $T_n^{(A2)} \xrightarrow{\mathcal{L}} \sup_{0\leqslant x\leqslant 1}\sum_{1\leqslant l\leqslant d} B_l^2(x)$.

(b) *The following statistics are suitable to detect epidemic change alternatives.*

$$T_n^{(B1)} = \frac{1}{n^3}\sum_{1\leqslant k_1<k_2\leqslant n}\mathbf{S}_n(k_1/n,k_2/n)^T\,\widehat{\Sigma}^{-1}\mathbf{S}_n(k_1/n,k_2/n),$$

$$T_n^{(B2)} = \max_{1\leqslant k_1<k_2\leqslant n}\frac{1}{n}\mathbf{S}_n(k_1/n,k_2/n)^T\,\widehat{\Sigma}^{-1}\mathbf{S}_n(k_1/n,k_2/n),$$

$$\text{where } \mathbf{S}_n(x,y) = \mathbf{S}_n(y) - \mathbf{S}_n(x) = \sum_{nx<j\leqslant ny}\left(\widehat{\boldsymbol{\eta}}_j - \frac{1}{n}\sum_{i=1}^{n}\widehat{\boldsymbol{\eta}}_i\right).$$

Under H_0 it holds:

(i) $T_n^{(B1)} \xrightarrow{\mathcal{L}} \sum_{1\leqslant l\leqslant d}\iint_{0\leqslant x<y\leqslant 1}(B_l(x) - B_l(y))^2\,dx\,dy$

(ii) $T_n^{(B2)} \xrightarrow{\mathcal{L}} \sup_{0\leqslant x<y\leqslant 1}\sum_{1\leqslant l\leqslant d}(B_l(x) - B_l(y))^2$.

Remark 3.2. For the above test statistics estimators of the long-run covariance matrix Σ are needed. Usually, estimators are of the following type:

$$\widehat{\Sigma} = \sum_{|h|\leqslant b_n} w_q(h/b_n)\widehat{\Gamma}(h),$$

for some appropriate weight function w_q and bandwidth b_n where

$$\widehat{\Gamma}(h) = \frac{1}{n}\sum_{j=1}^{n-h}\widehat{\boldsymbol{\eta}}_j\widehat{\boldsymbol{\eta}}_{j+h}^T, \quad h \geqslant 0, \qquad \widehat{\Gamma}(h) = \widehat{\Gamma}(-h),\ h < 0.$$

Hörmann and Kokoszka [14] prove consistency of this estimator for weakly dependent data. Politis [23] proposed to use different bandwidths for each entry of the matrix in addition to an automatic bandwidth selection procedure for the class of flat-top weight functions. Generally, in change point analysis it is advisable to adapt the estimators to take a possible change point into account to improve the power of the test. For details in the univariate situation we refer to Hušková and Kirch [17]. More details on problems and solutions for this estimator in the present situation for a functional magnetic resonance imaging (fMRI) data set can be found in Aston and Kirch [3].

3.2. Behavior under alternatives

In this section, we turn to the behavior of the test under the alternative hypothesis. We assume in the remainder of Sections 3.2 and 3.3 the existence of an orthonormal system $\{w_{k,j}(\cdot), k = 1, \ldots, d\}, j = A, B$, such that the orthonormal estimators $\widehat{v}_k(\cdot), k = 1, \ldots, d$, fulfill under alternatives

$$\int (\widehat{v}_k(t) - s_k w_{k,j}(t))^2\, dt = o_P(1), \quad s_k = \operatorname{sgn}\left(\int w_{k,j}(t)\,\widehat{v}_k(t)\, dt\right). \tag{3.21}$$

Lemma 2.4 and Theorem 2.1 show that this holds under very general conditions for estimated principal components. Note that w_k usually depends on the type of alternative. For clarity we sometimes write $w_{k,A}$ in case of an AMOC alternative and $w_{k,B}$ in case of an epidemic change alternative.

Lemma 3.2. *Let $\mathcal{P}\cdot 1$ and (3.21) hold.*

(a) *Under the AMOC-alternative $H_1^{(A)}$, it holds*

$$\sup_{0\leqslant x\leqslant 1}\left|\frac{1}{n}\sum_{1\leqslant i\leqslant nx}\widehat{\eta}_{i,l} - \frac{\lfloor nx\rfloor}{n^2}\sum_{i=1}^{n}\widehat{\eta}_{i,l} - s_l\int \Delta(t)w_{l,A}(t)\, dt\; g_A(x)\right| \xrightarrow{P} 0,$$

where

$$g_A(x) = \begin{cases} x(1-\vartheta), & 0 < x \leqslant \vartheta, \\ \vartheta(1-x), & \vartheta < x < 1, \end{cases}$$

$\Delta(t) = \mu_1(t) - \mu_2(t)$ and $w_{l,A}$ are as in (3.21).

(b) *Under the epidemic change alternative $H_1^{(B)}$, it holds*

$$\sup_{0\leqslant x<y\leqslant 1}\left|\frac{1}{n}\sum_{\lfloor nx\rfloor<i\leqslant ny}\widehat{\eta}_{i,l} - \frac{\lfloor ny\rfloor - \lfloor nx\rfloor}{n^2}\sum_{i=1}^{n}\widehat{\eta}_{i,l} - s_l\int \Delta(t)w_{l,B}(t)\, dt\; g_B(x,y)\right| \xrightarrow{P} 0,$$

where $g_B(x,y) = g_B(y) - g_B(x)$ and

$$g_B(x) = \begin{cases} x(\vartheta_2 - \vartheta_1), & x \leqslant \vartheta_1, \\ (\vartheta_1 - x(1-\vartheta_2-\vartheta_1)), & \vartheta_1 < x \leqslant \vartheta_2, \\ (x-1)(\vartheta_2-\vartheta_1), & x > \vartheta_2, \end{cases}$$

$\Delta(t) = \mu_1(t) - \mu_2(t)$ and $w_{k,B}(\cdot)$ are as in (3.21).

From the lemma we can conclude that the above tests are consistent in all cases where the change is not orthogonal to the contaminated projection subspace.

Theorem 3.2. *Assume that the estimator $\widehat{\Sigma}$ is symmetric with eigenvalues $\xi_{j,n}, j = 1, \ldots, d$ such that there exists $c > 0$ with $P(\min_{j=1,\ldots,d} 1/\xi_{j,n} \geqslant c) \to 1$ (as $n \to \infty$). Let (3.21) hold, in addition to*

$$\int \Delta(t)w_{k,j}(t)\, dt \neq 0, \tag{3.22}$$

$\Delta(t) = \mu_1(t) - \mu_2(t)$, for some $k = 1, \ldots, d$ with $j = A$ for (a) and $j = B$ for (b).

(a) *Under the AMOC alternative* $H_1^{(A)}$ *it holds*

$$\text{(i) } T_n^{(A1)} \xrightarrow{P} \infty, \qquad \text{(ii) } T_n^{(A2)} \xrightarrow{P} \infty.$$

(b) *Under the epidemic change alternative* $H_1^{(B)}$ *it holds*

$$\text{(i) } T_n^{(B1)} \xrightarrow{P} \infty, \qquad \text{(ii) } T_n^{(B2)} \xrightarrow{P} \infty.$$

The assumptions on the estimator $\widehat{\Sigma}$ are for example fulfilled if $\widehat{\Sigma} \xrightarrow{P} \Sigma_A$ for some symmetric positive-definite matrix Σ_A. The theorem shows that (3.22) is the crucial point indicating whether a change is detectable or not. Here, it is useful to use the first d estimated principal components because a large enough change will switch the projection subspace in such a way that (3.22) is fulfilled and the change becomes detectable (cf. Theorem 4.1).

Remark 3.3. Using Lemma 3.2 one even obtains consistency of the tests for local changes with $\|\Delta_n(\cdot)\| = \|\mu_{1,n}(\cdot) - \mu_{2,n}(\cdot)\| \to 0$ but $\sqrt{n}|\int \Delta_n(t) w_{k,j}(t)\, dt| \to \infty$.

3.3. Estimation of the change point

In this section we consider estimators for the change point ϑ under the AMOC alternative resp. for ϑ_1 and ϑ_2 under the epidemic change alternative and discuss consistency as well as rates of convergence.

First consider the AMOC alternative. Let $\arg\max(a(x) : x) = \min(x : a(x) = \max_y a(y))$ and consider the estimator

$$\widehat{\vartheta} = \arg\max\left(\mathbf{S}_n^T(x)\widehat{\Sigma}^{-1}\mathbf{S}_n(x) : 0 \leqslant x \leqslant 1\right), \tag{3.23}$$

$$\text{where } \mathbf{S}_n(x) = (S_{n,1}(x), \ldots, S_{n,d}(x))^T, \qquad S_{n,l}(x) = \sum_{1 \leqslant i \leqslant nx} \left(\widehat{\eta}_{i,l} - \frac{1}{n}\sum_{j=1}^{n} \widehat{\eta}_{j,l}\right).$$

Consistency of this estimator for i.i.d. observations and principal component estimation has been obtained by Berkes et al. [7] and follows immediately from Lemma 3.2. Rates have been obtained by Aue et al. [4] in this situation and their proof can be extended to the dependent situation.

Theorem 3.3. *Assume that the AMOC model holds. Furthermore, let the assumptions of Theorem 3.2 hold, i.e. the change is detectable, in addition to* $\widehat{\Sigma} \xrightarrow{P} \Sigma_A$ *for some symmetric positive-definite* Σ_A.

(a) *Then, the estimator* $\widehat{\vartheta}$ *is consistent, i.e.*

$$\widehat{\vartheta} - \vartheta = o_P(1).$$

(b) *If additionally (2.7) holds for* $\eta_{i,l}^{(A)} = \int Y_i(t)\, w_{k,A}(t)\, dt$, $l = 1, \ldots, d$, *then*

$$n(\widehat{\vartheta} - \vartheta) = O_P(1).$$

Aue et al. [4] additionally obtain the limit distribution of $n(\widehat{\vartheta} - \vartheta)$ in case of i.i.d. data showing that the rate in Theorem 3.3(b) cannot be improved.

In case of an epidemic change alternative we consider the estimator

$$(\widehat{\vartheta}_1, \widehat{\vartheta}_2) = \arg\max\left(\mathbf{S}_n^T(x, y)\widehat{\Sigma}_n^{-1}\mathbf{S}_n(x, y) : 0 \leqslant x < y \leqslant 1\right), \tag{3.24}$$
$$\text{where } \mathbf{S}_n(x, y) = \mathbf{S}_n(y) - \mathbf{S}_n(x)$$

and $(x_1, y_1) = \arg\max(Z(x, y) : 0 \leqslant x < y \leqslant 1)$ iff $x_1 = \min(0 \leqslant x < 1 : Z(x, y) = \max_{0 \leqslant s < t \leqslant 1} Z(s, t)$ for some $y)$ and $y_1 = \max(y > x_1 : Z(x_1, y) = \max_{0 \leqslant s < t \leqslant 1} Z(s, t))$.

Theorem 3.4. *Assume that the epidemic change model holds. Furthermore, let the assumptions of Theorem 3.2 hold in addition to* $\widehat{\Sigma} \xrightarrow{P} \Sigma_A$ *for some symmetric positive-definite* Σ_A.

(a) *Then, the estimator* $(\widehat{\vartheta}_1, \widehat{\vartheta}_2)$ *is consistent, i.e.*

$$(\widehat{\vartheta}_1 - \vartheta_1, \widehat{\vartheta}_2 - \vartheta_2)^T = o_P(1).$$

(b) *If additionally (2.7) holds for* $\eta_{i,l}^{(B)} = \int Y_i(t)\, w_{k,B}(t)\, dt$, $l = 1, \ldots, d$, *then*

$$n(\widehat{\vartheta}_1 - \vartheta_1, \widehat{\vartheta}_2 - \vartheta_2)^T = O_P(1).$$

4. Principal component analysis for subspace selection

In this section we return to a subspace selection based on principal components and show that it is rather useful in the context of mean change models. In Sections 3.2 and 3.3 we have seen that the tests have asymptotic power one and the estimators are consistent if the change $\Delta(\cdot)$ is not orthogonal to the contaminated projection subspace, which depends directly on both the change and time of change. The following theorem allows a characterization of detectable changes in terms of the non-contaminated projection subspace and even more importantly shows that the change has a tendency to influence the contaminated projection subspace in such a way that it becomes detectable.

Theorem 4.1. (a) *Let $\{v_l, l = 1, \ldots, d\}$ be the eigenfunctions belonging to the largest d eigenvalues of $c(t, s)$ as well as $\{w_l, l = 1, \ldots, d\}$ the eigenfunctions belonging to the largest d eigenvalues of $k(t, s)$ as in (2.14).*
For $\Delta(t) = \mu_1(t) - \mu_2(t)$ it holds

$$\int \Delta(t) v_l(t)\, dt \neq 0 \quad \text{for some } l = 1, \ldots, d$$
$$\Longrightarrow \int \Delta(t) w_l(t)\, dt \neq 0 \quad \text{for some } l = 1, \ldots, d.$$

This shows, that any change that is not orthogonal to the non-contaminated subspace is detectable.
(b) *Let $\Delta_D(t) = D\,\Delta(t)$, $\int \Delta^2(t)\, dt \neq 0$. Then, there exists $D_0 > 0$ such that*

$$\int \Delta_D(t) w_{1,D}(t)\, dt \neq 0$$

for all $|D| \geqslant D_0$, where $w_{1,D}$ is the eigenfunction belonging to the largest eigenvalue of the contaminated covariance kernel $k_D(t, s) = c(t, s) + \theta(1-\theta)\Delta_D(t)\Delta_D(s)$. This shows, that any large enough change is detectable. It even holds (as $D \to \infty$)

$$\left\| w_{1,D}(\cdot) - s\frac{\Delta(\cdot)}{\|\Delta(\cdot)\|} \right\| \to 0, \quad s = \operatorname{sgn} \int \Delta(t)\, w_{1,D}(t)\, dt.$$

Remark 4.1. Theorem 4.1(a) shows that we are able to detect at least all changes that are not orthogonal to the non-contaminated subspace spanned by the first d principal components. Part (b) shows that frequently changes can be detected even if they are orthogonal to the non-contaminated ON-System. The reason is that large mean changes lead to a larger variability of the empirical covariance function and thus the contaminated covariance function $k(t, s) = c(t, s) + \theta(1 - \theta)\Delta(t)\Delta(s)$ in the components that are not orthogonal to the change, while not changing the variability in the components that are orthogonal. In the following example such a change in the subspace takes place: Let $\{b_j : j \geqslant 1\}$ be an orthonormal basis of the continuous functions on $\mathcal{Z}$. Furthermore X, Y are i.i.d. $N(0, 1)$, and $Y(t) = 2Xb_1(t) + Yb_2(t)$. Obviously $c(t, s)$ has the eigenvalues 4 with eigenfunction b_1 as well as the eigenvalue 1 with eigenfunction b_2 in addition to the eigenvalue 0. With the notation of (2.14) let $\theta = 1/2$ and consider $\Delta(t) = 4b_2(t)$ which for $d = 1$ is obviously orthogonal to b_1, but it is easy to see that the eigenvalues of $k(t, s)$, are now 5 corresponding to b_2 and 4 corresponding to b_1 in addition to the eigenvalue 0. This shows that the mean change is no longer orthogonal to the space spanned by the eigenfunction corresponding to the largest eigenvalue, which is the one spanned by b_2.

An immediate corollary to Theorem 4.1 also gives rise to a surprising fact for multivariate data. PCA is well known to work poorly as a representation of the data when the covariance matrix of the multivariate observations is close to a multiple of the identity matrix. In fact, the scree plot will be constant in nature in the case when the covariance is an exact multiple of the identity implying that there is no effective sparse representation of the data. As a contrast, by the proof of the theorem above, it is optimal for detecting a change point if the uncontaminated covariance matrix is a multiple of the identity matrix. In this case, choosing only a single principal component from the contaminated covariance will guarantee that the power of detection is asymptotically one. Thus PCA based change point detection (for either epidemic or AMOC) works best when PCA itself works worst for the uncontaminated system regardless of the direction of the change.

This fact also translates over to functional data, but because the eigenvalues are square summable, the degenerate case will not occur. However, in situations where the eigenvalues decay very rapidly in the uncontaminated case, changes orthogonal to the eigenfunctions corresponding to the largest uncontaminated eigenvalues will be required to be bigger if they are supposed to be detectable than in situations with more slowly decreasing eigenvalues.

5. Proofs

5.1. Proofs of Section 2

Proof of Lemma 2.1. If $\{Y_j(\cdot)\}$ is $L^2 - m$-approximable, then assertion (a) follows from Hörmann and Kokoszka [14, Corollary 4.1], while (b) follows from Theorem A.2.

If $\{Y_j(\cdot)\}$ is strong mixing then by definition $\{\eta_{i,j} : i = 1, \ldots, n\}$ is strong-mixing with the same rate. Furthermore, the $(2+\delta)$-moment of $\boldsymbol{\eta}_i$ exists by an application of the Hölder inequality. Assertion (a) and (b) follow from Theorem 4 in Kuelbs and Philipp [20]. Note that the reverse of a strong mixing sequence is again strong mixing, which implies the reverse Hájek-Rényi inequality. □

Proof of Lemma 2.2. The assertion for strong mixing sequences follows from the invariance principle of Kuelbs and Philipp [20, Theorem 4] in addition to the standard Hájek-Rényi-inequality for independent random variables.

For $L^4 - m$-approximable sequences we will show that for some constant $C > 0$

$$\mathrm{E}\left(\sum_{j=l}^{k} \xi_j\right)^4 \leqslant C(k-l+1)^2, \tag{5.25}$$

from which the standard as well as reverse Hájek-Rényi-type inequality follows from a generalization of the results obtained in Lavielle and Moulines [21], Móricz et al. [22] as in Kirch [19, Theorem B.1]. By stationarity of $\{\xi_j\}$ it is sufficient to show that

$$\mathrm{E}\left(\sum_{j=1}^{k} \xi_j\right)^4 \leqslant Ck^2.$$

Because $\{\xi_i\}$ is centered, it holds

$$\mathrm{E}\left(\sum_{j=1}^{k} \xi_j\right)^4 = \sum_{l_1,\ldots,l_4=1}^{k} \mathrm{cov}(\xi_{l_1}\xi_{l_2}, \xi_{l_3}\xi_{l_4}) + \left(\sum_{l_1,l_2=1}^{k} \mathrm{cov}(\xi_{l_1}, \xi_{l_2})\right)^2 = \sum_{l_1,\ldots,l_4=1}^{k} \mathrm{cov}(\xi_{l_1}\xi_{l_2}, \xi_{l_3}\xi_{l_4}) + O(k^2)$$

by the absolute summability of the auto-covariance function of $L^4 - m$-approximable sequences (confer Lemma 4.1 in Hörmann and Kokoszka [14]). Analogous arguments as in the proof of Theorem 4.1 in Hörmann and Kokoszka [14] taking (2.6) into account show that the first term is also bounded by Ck^2 finishing the proof of (5.25). □

Proof of Lemma 2.3. By ergodicity and stationarity the following law of large numbers holds (cf. e.g. Ranga Rao [24])

$$\left\| \frac{1}{n} \sum_{i=1}^{n} Y_i(\cdot) \right\|_{L^2(\mathcal{Z})} \to 0 \quad a.s. \tag{5.26}$$

Using this result, assertion (a) follows analogously to Berkes et al. [7, proof of Lemma 1].

Assertion (b) for $L^p - m$-approximable sequences has been proven in Hörmann and Kokoszka [14, Theorem 3.1]. The proof for mixing sequences is very similar, where we use the version of Davydov's covariance inequality for Hilbert space valued random variables due to Dehling and Philipp [10, Lemma 2.2] ($t^{-1} + r^{-1} + s^{-1} = 1$):

$$|\mathrm{E}\langle Y_1(\cdot), Y_{1+h}(\cdot)\rangle - \langle \mathrm{E}Y_1(\cdot), \mathrm{E}Y_{1+h}(\cdot)\rangle| \leqslant 15\alpha_h^{1/t} \left(\mathrm{E}\|Y_1(\cdot)\|^r\right)^{1/r} \left(\mathrm{E}\|Y_{1+h}(\cdot)\|^s\right)^{1/s}, \tag{5.27}$$

where $\langle \cdot, \cdot \rangle = \langle \cdot, \cdot \rangle_{\mathcal{Z}}$ is the scalar product on $L^2(\mathcal{Z})$. It holds

$$\iint (\widehat{c}_n(t,s) - c(t,s))^2 \, dt\, ds \leqslant 2 \iint \left(\frac{1}{n}\sum_{i=1}^{n} Y_i(t)Y_i(s) - \mathrm{E}Y_1(t)Y_1(s)\right)^2 dt\, ds + 2\left\| \frac{1}{n}\sum_{i=1}^{n} Y_i(\cdot)\right\|^4.$$

$Z_i(t,s) = Y_i(t)Y_i(s) \in L^2(\mathcal{Z} \times \mathcal{Z})$ is strong mixing with mixing rate α_h. Some calculations and (5.27) yield

$$\begin{aligned} n\mathrm{E}\iint \left(\frac{1}{n}\sum_{i=1}^{n} Y_i(t)Y_i(s) - \mathrm{E}Y_1(t)Y_1(s)\right)^2 dt\, ds &= \sum_{|h|<n} \left(1 - \frac{|h|}{n}\right) (\mathrm{E}\langle Z_1, Z_{1+h}\rangle_{\mathcal{Z}\times\mathcal{Z}} - \langle \mathrm{E}Z_1, \mathrm{E}Z_{1+h}\rangle_{\mathcal{Z}\times\mathcal{Z}}) \\ &\leqslant c\left(\mathrm{E}\|Z_1\|_{\mathcal{Z}\times\mathcal{Z}}^{2+\delta/2}\right)^{\frac{2}{2+\delta/2}} \sum_{h\geqslant 1} \alpha_h^{\frac{\delta}{4+\delta}} \leqslant c'\left(\mathrm{E}\|Y_1(\cdot)\|_{\mathcal{Z}}^{4+\delta}\right)^{\frac{4}{4+\delta}} < \infty, \end{aligned}$$

for some constants $c, c' > 0$. Hence $\iint \left(\frac{1}{n}\sum_{i=1}^{n} Y_i(t)Y_i(s) - \mathrm{E}Y_1(t)Y_1(s)\right)^2 dt\, ds = O_P(n^{-1})$. Analogously one obtains

$$\left\| \frac{1}{n}\sum_{i=1}^{n} Y_i(\cdot)\right\|^2 = O_P(n^{-1}). \quad \square$$

Proof of Lemma 2.4. The assertion follows analogously to Berkes et al. [7, proof of Lemma 1], on using (5.26). □

Proof of Theorem 2.1. The assertion follows immediately from the assumptions and Lemmas 4.2 and 4.3 of Bosq [8]. □

5.2. Proofs of Section 3

Most of the proofs in this section follow the ideas of proofs given in either Berkes et al. [7] (for the proofs of Sections 3.1 and 3.2) or Aue et al. [4] (for the proofs of Section 3.3) in the AMOC situation for i.i.d. functional data using a subspace obtained from principal components analysis. This independent setup allows to consider only the simpler situation, where $\mathbf{\Sigma}$ is a diagonal matrix. In the dependent situation considered here $\mathbf{\Sigma}$ is no longer guaranteed to be diagonal. Furthermore, in Section 2 we identified the main properties a weakly dependent time series as well as the projection subspace need to have in order for the proofs of the i.i.d. AMOC situation still to be valid allowing us to generalize those results in Section 3 to the dependent AMOC but also epidemic change situation, which we prove in this section.

Proof of Lemma 3.1. First note that under H_0

$$\widehat{\eta}_{i,l} - \bar{\widehat{\eta}}_l = \check{\eta}_{i,l} - \bar{\check{\eta}}_l,$$

where (e.g.) $\bar{\widehat{\eta}}_l = \frac{1}{n}\sum_{i=1}^n \widehat{\eta}_{i,l}$. Furthermore

$$\begin{aligned}\sup_{0<x<1}\left|\frac{1}{\sqrt{n}}\sum_{1\leqslant i\leqslant nx} s_l\check{\eta}_{i,l} - \frac{1}{\sqrt{n}}\sum_{1\leqslant i\leqslant nx}\eta_{i,l}\right| &= \sup_{0<x<1}\left|\int\left(\frac{1}{n}\sum_{1\leqslant i\leqslant nx} Y_i(t)\right)\sqrt{n}\,(s_l\widehat{v}_l(t) - v_l(t))\,dt\right| \\ &\leqslant \sup_{0<x<1}\left(\int\left(\frac{1}{n}\sum_{1\leqslant i\leqslant nx} Y_i(t)\right)^2\right)^{1/2}\left(n\int(s_l\widehat{v}_l(t) - v_l(t))^2\right)^{1/2} = o_P(1),\end{aligned}$$

where the last line follows by (5.26). Hence by standard arguments

$$\sup_{0<x<1}\int\left(\frac{1}{n}\sum_{1\leqslant i\leqslant nx} Y_i(t)\right)^2 dt = o_P(1), \tag{5.28}$$

as the second factor is $O_P(1)$ by (3.15) and the fact that $s_l^2 = 1$, $s_l^{-1} = s_l$.

The assertion now follows from (2.5). □

Proof of Theorem 3.1. The assertions of the theorem follow immediately from Lemma 3.1 and the fact that $s_l^2 = 1$. □

Proof of Lemma 3.2. Concerning (a), by (3.17) and (5.28) it holds uniformly in $x \leqslant \vartheta$

$$\begin{aligned}\frac{1}{n}\sum_{1\leqslant i\leqslant nx}\widehat{\eta}_{i,l} - \frac{\lfloor nx\rfloor}{n^2}\sum_{i=1}^n\widehat{\eta}_{i,l} &= x(1-\vartheta)\left(\int\mu_1(t)\widehat{v}_l(t)\,dt - \int\mu_2(t)\widehat{v}_l(t)\,dt\right) + o_P(1) \\ &= x(1-\vartheta)s_l\int(\mu_1(t) - \mu_2(t))w_{l,A}(t)\,dt + o_P(1),\end{aligned}$$

where the last line follows from (3.21). Analogously one obtains uniformly in $x > \vartheta$

$$\frac{1}{n}\sum_{1\leqslant i\leqslant nx}\widehat{\eta}_{i,l} - \frac{\lfloor nx\rfloor}{n^2}\sum_{i=1}^n\widehat{\eta}_{i,l} = \vartheta(1-x)s_l\int(\mu_1(t) - \mu_2(t))w_{l,A}(t)\,dt + o_P(1),$$

which finishes the proof of (a).

Concerning (b) note that for the epidemic change alternative one gets analogously uniformly in x

$$\frac{1}{n}\sum_{1\leqslant i\leqslant nx}\widehat{\eta}_{i,l} - \frac{\lfloor nx\rfloor}{n^2}\sum_{i=1}^n\widehat{\eta}_{i,l} = o_P(1) + \begin{cases} x(\vartheta_2-\vartheta_1)\int\Delta(t)w_{l,B}(t)\,dt, & x\leqslant\vartheta_1, \\ (\vartheta_1 - x(1-\vartheta_2-\vartheta_1))\int\Delta(t)w_{l,B}(t)\,dt, & \vartheta_1 < x\leqslant\vartheta_2, \\ (x-1)(\vartheta_2-\vartheta_1)\int\Delta(t)w_{l,B}(t)\,dt, & x>\vartheta_2,\end{cases}$$

yielding the assertion. □

Proof of Theorem 3.2. Consider $\mathbf{d} = (\int\Delta(t)w_{1,A}(t)\,dt, \ldots, \int\Delta(t)w_{d,A}(t)\,dt)^T$ with $\mathbf{d}^T\mathbf{d} > 0$ by assumption. Due to the assumptions on $\widehat{\Sigma}$ there exists a unitary matrix U such that

$$\widehat{\Sigma}^{-1} = U^T\Xi^{-1}U, \qquad \Xi^{-1} = (\xi_{j,n}^{-1}1_{\{j=l\}})_{j,l=1,\ldots,d}.$$

Moreover, by assumption there exists $c > 0$ such that $P(\min_{j=1,\dots,d} 1/\xi_{j,n} \geqslant c) \to 1$ and on this set $\mathbf{d}^T \widehat{\Sigma}^{-1}\mathbf{d} \geqslant c\,\mathbf{d}^T\mathbf{d}$, hence with $D = c\mathbf{d}^T\mathbf{d} > 0$

$$P\left(\mathbf{d}^T \widehat{\Sigma}^{-1}\mathbf{d} \geqslant D\right) = 1 + o(1).$$

By Lemma 3.2 we obtain

$$T_n^{(A1)} = n\left(\mathbf{d}^T \widehat{\Sigma}^{-1}\mathbf{d}\left(\int_0^1 g_A^2(x)\,dx + o(1)\right) + o_P(1)\right) \xrightarrow{P} \infty$$

and analogously $T_n^{(B1)} \xrightarrow{P} \infty$. Furthermore

$$T_n^{(A2)} \geqslant n\left(\mathbf{d}^T \widehat{\Sigma}^{-1}\mathbf{d}\, g_A^2(\vartheta) + o_P(1)\right) \xrightarrow{P} \infty$$

and analogously $T_n^{(B2)} \xrightarrow{P} \infty$. □

Proof of Theorem 3.3. Lemma 3.2 implies

$$\sup_{0\leqslant x\leqslant 1}\left|\frac{1}{n^2}\mathbf{S}_n^T(x)\widehat{\Sigma}^{-1}\mathbf{S}_n(x) - g_A^2(x)\mathbf{d}^T\Sigma_A^{-1}\mathbf{d}\right| = o_P(1),$$

where $\mathbf{d} = (\int \Delta(t)w_{1,A}(t)\,dt, \dots, \int \Delta(t)w_{d,A}(t)\,dt)^T$. Since $\mathbf{d}^T\Sigma_A^{-1}\mathbf{d} > 0$ and $g_A^2(\cdot)$ has a unique maximum at $x = \vartheta$ and is continuous, assertion (a) follows by standard arguments, noting that $\widehat{\vartheta} = \arg\max \mathbf{S}_n^T(x)\widehat{\Sigma}^{-1}\mathbf{S}_n(x)/n^2$.

Note that $\widehat{\vartheta}$ is obtained as the arg max of $Q_n(x) := \mathbf{S}_n(x)^T\widehat{\Sigma}^{-1}\mathbf{S}_n(x)$. This is equivalent to $\widehat{\vartheta} = \widehat{k}/n$ and $\widehat{k} = \arg\max(Q_n(k/n) - Q_n(\lfloor\vartheta n\rfloor/n) : k = 1, \dots, n)$. The key to the proof is now the following decomposition for $k \leqslant k^\circ := \lfloor n\vartheta\rfloor$ which generalizes equation (4.1) in [4] for situations where $\widehat{\Sigma}$ has no diagonal shape. Since for a symmetric matrix C it holds $(a-b)^TC(a+b) = a^TCa - b^TCb$ we get by (3.17)

$$Q_n(k/n) - Q_n(k^\circ/n) = (\mathbf{A}_k^{(1)} + \widehat{\mathbf{d}}B_k^{(1)})^T\widehat{\Sigma}^{-1}(\mathbf{A}_k^{(2)} + \widehat{\mathbf{d}}B_k^{(2)}), \tag{5.29}$$

for $\check{\eta}$ as in (3.16), $\mathbf{A}_k^{(j)} = (A_{k,1}^{(j)}, \dots, A_{k,d}^{(j)})^T$, $j = 1, 2$, where

$$A_{k,l}^{(1)} = -\sum_{i=k+1}^{k^\circ}\check{\eta}_{i,l} - \frac{k-k^\circ}{n}\sum_{i=1}^{n}\check{\eta}_{i,l}, \qquad A_{k,l}^{(2)} = \sum_{i=1}^{k}\check{\eta}_{i,l} + \sum_{i=1}^{k^\circ}\check{\eta}_{i,l} - \frac{k+k^\circ}{n}\sum_{i=1}^{n}\check{\eta}_{i,l},$$

$\widehat{\mathbf{d}} = (\widehat{d}_1, \dots, \widehat{d}_d)^T$ with $\widehat{d}_l = \int(\mu_1(t) - \mu_2(t))\widehat{v}_l(t)\,dt$ and

$$B_k^{(1)} = (k-k^\circ)\frac{n-k^\circ}{n}, \qquad B_k^{(2)} = (k+k^\circ)\frac{n-k^\circ}{n}.$$

We will first show that the following term becomes arbitrarily small for $N \to \infty$:

$$P(n\widehat{\vartheta} \leqslant n\vartheta - N) = P(\widehat{k} \leqslant k^\circ - N).$$

To this end we consider $k \leqslant k^\circ - N$. We show that $B_k^{(1)}B_k^{(2)}\,\mathbf{d}^T\widehat{\Sigma}^{-1}\mathbf{d}$ is the dominating term in the decomposition (5.29). Let

$$L_{n,k} = -(k^\circ - k)(k + k^\circ)\left(\frac{n-k^\circ}{n}\right)^2,$$

$$\text{i.e. } \left|L_{n,k}\right| \geqslant (k^\circ - k)n\left(\vartheta(1-\vartheta)^2 + o(1)\right). \tag{5.30}$$

Then,

$$B_k^{(1)}B_k^{(2)}\,\widehat{\mathbf{d}}^T\widehat{\Sigma}^{-1}\widehat{\mathbf{d}} = L_{n,k}\,\widehat{\mathbf{d}}^T\widehat{\Sigma}^{-1}\widehat{\mathbf{d}} = L_{n,k}(\mathbf{d}^T\Sigma_A^{-1}\mathbf{d} + o_P(1)), \tag{5.31}$$

since by assumption $\widehat{\Sigma} \xrightarrow{P} \Sigma_A$ and by Theorem 2.1 and the Cauchy–Schwarz inequality it holds for $d_l = \int(\mu_1(t) - \mu_2(t))w_{l,A}(t)\,dt$,

$$\left|\widehat{d}_l - s_l d_l\right| \leqslant \|\mu_1(\cdot) - \mu_2(\cdot)\|\,\|\widehat{v}_l - s_l w_{l,A}\| = o_P(1).$$

Similarly

$$\max_{1\leqslant k\leqslant k^\circ - N}\frac{|\widehat{d}_l B_k^{(1)}|}{k^\circ - k} = O_P(1), \qquad \max_{1\leqslant k\leqslant k^\circ - N}\frac{|\widehat{d}_l B_k^{(2)}|}{n} = O_P(1). \tag{5.32}$$

By stationarity it holds

$$\max_{1\leqslant k<k^\circ}\left\|\frac{1}{k^\circ-k}\sum_{i=k+1}^{k^\circ}Y_i(\cdot)\right\| \overset{\mathcal{L}}{=} \max_{1\leqslant k<k^\circ}\left\|\frac{1}{k}\sum_{i=1}^{k}Y_{-i}(\cdot)\right\| = O_P(1),$$

where the last line follows by the (a.s.) ergodic theorem because the reversed process $\{Y_{-i}\}$ remains stationary. Hence we get by (3.21)

$$\left|\max_{1\leqslant k\leqslant k^\circ-N}\frac{1}{k^\circ-k}\left|\sum_{i=k+1}^{k^\circ}\check{\eta}_{i,l}\right| - \max_{1\leqslant k\leqslant k^\circ-N}\frac{1}{k^\circ-k}\left|\sum_{i=k+1}^{k^\circ}\eta_{i,l}^{(A)}\right|\right| \leqslant \left\|\widehat{v}_l(t)-s_k w_{l,A}(t)\right\| \max_{1\leqslant k<k^\circ}\left\|\frac{1}{k^\circ-k}\sum_{i=k+1}^{k^\circ}Y_i(\cdot)\right\| = o_P(1)$$

as well as by (2.7)

$$\max_{1\leqslant k\leqslant k^\circ-N}\frac{1}{k^\circ-k}\left|\sum_{i=k+1}^{k^\circ}\eta_{i,l}^{(A)}\right| \leqslant \alpha(N)^{-1}O_P(1),$$

which becomes arbitrarily small as $N\to\infty$. This shows

$$\max_{1\leqslant k\leqslant k^\circ-N}\frac{|A_{k,l}^{(1)}|}{k^\circ-k} \leqslant \alpha(N)^{-1}O_P(1)+o_P(1). \tag{5.33}$$

Since by the Cauchy–Schwarz inequality and (5.28)

$$\max_{1\leqslant k\leqslant n}\frac{1}{n}\left|\sum_{i=1}^{k}\eta_{i,l}^{(A)}\right| \leqslant \|w_{l,A}\| \max_{1\leqslant k\leqslant n}\left\|\frac{1}{n}\sum_{i=1}^{k}Y_i(\cdot)\right\| = o_P(1),$$

we get

$$\max_{1\leqslant k\leqslant k^\circ-N}\frac{|A_{k,l}^{(2)}|}{n} = o_P(1). \tag{5.34}$$

From $\widehat{\Sigma}\xrightarrow{P}\Sigma_A$, (5.30), (5.33) and (5.34) we can conclude

$$\max_{1\leqslant k\leqslant k^\circ-N}\frac{\left|\mathbf{A}_k^{(1)T}\widehat{\Sigma}^{-1}\mathbf{A}_k^{(2)}\right|}{L_{n,k}} = O_P(1)\max_{l=2,\ldots,d}\max_{1\leqslant k\leqslant k^\circ-N}\frac{|A_{k,l}^{(1)}|}{k^\circ-k}\max_{1\leqslant k\leqslant k^\circ-N}\frac{|A_{k,l}^{(2)}|}{n} = o_P(1)$$

and similarly using additionally (5.32) it holds

$$\max_{1\leqslant k\leqslant k^\circ-N}\frac{\left|\mathbf{A}_k^{(1)T}\widehat{\Sigma}^{-1}\mathbf{d}B_k^{(2)}\right|}{L_{n,k}} \leqslant \alpha(N)^{-1}O_P(1)+o_P(1), \qquad \max_{1\leqslant k\leqslant k^\circ-N}\frac{\left|B_k^{(1)}\mathbf{d}^T\widehat{\Sigma}^{-1}\mathbf{A}_k^{(2)}\right|}{L_{n,k}} = o_P(1).$$

This in addition to (5.31) and

$$\max_{k\leqslant k^\circ-N}L_{n,k} \leqslant -Nk^\circ\left(1-\frac{k^\circ}{n}\right)^2 < 0 \tag{5.35}$$

we finally obtain

$$\begin{aligned}
P(n\widehat{\vartheta}\leqslant n\vartheta-N) &= P(\widehat{k}\leqslant k^\circ-N) = P\left(\max_{k\leqslant k^\circ-N}Q_n\left(\frac{k}{n}\right)-Q_n\left(\frac{k^\circ}{n}\right)\geqslant \max_{k>k^\circ-N}Q_n\left(\frac{k}{n}\right)-Q_n\left(\frac{k^\circ}{n}\right)\right)\\
&\leqslant P\left(\max_{k\leqslant k^\circ-N}Q_n\left(\frac{k}{n}\right)-Q_n\left(\frac{k^\circ}{n}\right)\geqslant 0\right) \leqslant P\left(\left(\mathbf{d}^T\Sigma_A^{-1}\mathbf{d}+o_P(1)+\alpha(N)^{-1}O_P(1)\right)\max_{k\leqslant k^\circ-N}L_{n,k}\geqslant 0\right)\\
&\leqslant P\left(\alpha(N)^{-1}O_P(1)\geqslant \mathbf{d}^T\Sigma_A^{-1}\mathbf{d}+o_P(1)\right) \leqslant P(O_P(1)\geqslant \alpha(N)\,\mathbf{d}^T\Sigma_A^{-1}\mathbf{d})+o_P(1),
\end{aligned}$$

which becomes arbitrarily small if $N\to\infty$, since by assumption $\mathbf{d}^T\Sigma_A^{-1}\mathbf{d}>0$.

Analogous arguments for $k\geqslant k^\circ+N$ show that $P(n\widehat{\vartheta}\geqslant n\vartheta+N)$ becomes arbitrarily small as $N\to\infty$, which finishes the proof. In fact, for $k\geqslant k^\circ+N$ the arguments to obtain the analogue of (5.33) simplify because no time inversion is needed. □

Proof of Theorem 3.4. Note that $g_B(x)$ is continuous and has a unique maximum at $x=\vartheta_1$ and a unique minimum at $x=\vartheta_2$, hence $g_B(x,y)=g_B(y)-g_B(x)$ is continuous and has a unique (for $x<y$) maximum at $(\vartheta_1,\vartheta_2)$. Then, the proof of (a) is completely analogous to the proof of (a) of Theorem 3.3.

The proof of (b) is close to the proof of Theorem 3.3(b), we therefore only sketch it here. Let $Q_n(k_1, k_2) := \mathbf{S}_n(k_1/n, k_2/n)^T \widehat{\Sigma}^{-1} \mathbf{S}_n(k_1/n, k_2/n)$, then $(\widehat{k}_1, \widehat{k}_2) = \arg\max(Q_n(k_1, k_2) - Q_n(k_1^\circ, k_2^\circ))$ where $\widehat{\vartheta}_j = \widehat{k}_j/n$.

Note that by an analogous expression to (3.17) it holds ($k_j^\circ := \lfloor n\vartheta_j \rfloor$, $a_+ = \max(a, 0)$)

$$S_{n,l}\left(\frac{k}{n}\right) = \sum_{j=1}^{k} \check{\eta}_{j,l} - \frac{k}{n}\sum_{j=1}^{n} \check{\eta}_{j,l} + \widehat{d}_l \left(\frac{k(k_2^\circ - k_1^\circ)}{n} - \left(\min(k, k_2^\circ) - k_1^\circ\right)_+\right),$$

where $\widehat{d}_l = \int (\mu_1(t) - \mu_2(t))\widehat{v}_l\, dt$ and $\widehat{\mathbf{d}} = (\widehat{d}_1, \ldots, \widehat{d}_n)^T$. Analogously to (5.29) we get

$$\begin{aligned}
&Q_n(k_1, k_2) - Q_n(k_1^\circ, k_2^\circ) \\
&= \left(\mathbf{S}_n\left(\frac{k_2}{n}\right) - \mathbf{S}_n\left(\frac{k_2^\circ}{n}\right) - \mathbf{S}_n\left(\frac{k_1}{n}\right) + \mathbf{S}_n\left(\frac{k_1^\circ}{n}\right)\right)^T \widehat{\Sigma}^{-1} \left(\mathbf{S}_n\left(\frac{k_2}{n}\right) - \mathbf{S}_n\left(\frac{k_1}{n}\right) + \mathbf{S}_n\left(\frac{k_2^\circ}{n}\right) - \mathbf{S}_n\left(\frac{k_1^\circ}{n}\right)\right) \\
&= (\mathbf{A}^{(1)}_{k_1,k_2} - \widehat{\mathbf{d}}\, B^{(1)}_{k_1,k_2})^T \widehat{\Sigma}^{-1} (\mathbf{A}^{(2)}_{k_1,k_2} - \widehat{\mathbf{d}}\, B^{(2)}_{k_1,k_2}),
\end{aligned}$$

where $\mathbf{A}^{(j)}_{k_1,k_2} = (A^{(j)}_{k_1,k_2,1}, \ldots, A^{(j)}_{k_1,k_2,d})^T$, $j = 1, 2$, and

$$\begin{aligned}
A^{(1)}_{k_1,k_2,l} &= z_2 \sum_{j=m_2+1}^{M_2} \check{\eta}_{j,l} - z_1 \sum_{j=m_1+1}^{M_1} \check{\eta}_{j,l} - \frac{k_2 - k_2^\circ}{n}\sum_{j=1}^{n} \check{\eta}_{j,l} + \frac{k_1 - k_1^\circ}{n}\sum_{j=1}^{n} \check{\eta}_{j,l}, \\
A^{(2)}_{k_1,k_2,l} &= \sum_{j=1}^{k_2} \check{\eta}_{j,l} - \sum_{j=1}^{k_1} \check{\eta}_{j,l} + \sum_{j=1}^{k_2^\circ} \check{\eta}_{j,l} - \sum_{j=1}^{k_1^\circ} \check{\eta}_{j,l} - \frac{k_2 - k_1 + k_2^\circ - k_1^\circ}{n}\sum_{j=1}^{n} \check{\eta}_{j,l}, \\
B^{(1)}_{k_1,k_2} &= (m_2 - k_1^\circ)_+ - (k_2^\circ - k_1^\circ) - (\min(k_1, k_2^\circ) - k_1^\circ)_+ - \frac{k_2^\circ - k_1^\circ}{n}(k_2 - k_2^\circ - k_1 + k_1^\circ), \\
B^{(2)}_{k_1,k_2} &= (m_2 - k_1^\circ)_+ - (\min(k_1, k_2^\circ) - k_1^\circ)_+ + (k_2^\circ - k_1^\circ) - \frac{k_2^\circ - k_1^\circ}{n}\left(k_2^\circ - k_1^\circ + k_2 - k_1\right)
\end{aligned}$$

with $z_j = 1$ if $k_j > k_j^\circ$ and $z_j = -1$ else, $m_j = \min(k_j, k_j^\circ)$, $M_j = \max(k_j, k_j^\circ)$.

We will show that the deterministic part is dominating as long as $\max(|k_1 - k_1^\circ|, |k_2 - k_2^\circ|) > N$. Here, the problem is that the maximum needs to be divided into six parts (instead of just two as in the proof of Theorem 3.3). Let

$$L_{n,k_1,k_2} := B^{(1)}_{k_1,k_2} B^{(2)}_{k_1,k_2}$$

In all six cases one can then show that analogously to (5.31)

$$\frac{\left|L_{n,k_1,k_2}\right|}{n\max(|k_1 - k_1^\circ|, |k_2 - k_2^\circ|)} \geqslant c + o(1) > 0 \tag{5.36}$$

as well as analogously to (5.35)

$$\max L_{n,k_1,k_2} < 0. \tag{5.37}$$

Due to limitations of space we only give the proof exemplary for the case where $0 \leqslant k_1 \leqslant k_1^\circ < k_2 \leqslant k_2^\circ$ and $\max(|k_1 - k_1^\circ|, |k_2 - k_2^\circ|) > N$. The other cases are not completely analogous but similar arguments can be used. In the above case we obtain

$$-B^{(1)}_{k_1,k_2} = (k_2^\circ - k_2)\left(1 - \frac{k_2^\circ - k_1^\circ}{n}\right) + (k_1^\circ - k_1)\frac{k_2^\circ - k_1^\circ}{n}, \tag{5.38}$$

and hence there exists $c_1 > 0$ such that

$$\frac{-B^{(1)}_{k_1,k_2}}{\max(|k_1 - k_1^\circ|, |k_2 - k_2^\circ|)} \geqslant c_1 + o(1).$$

Similarly

$$\begin{aligned}
B^{(2)}_{k_1,k_2} &= k_2 - k_1^\circ + k_2^\circ - k_1^\circ - \frac{k_2^\circ - k_1^\circ}{n}\left(k_2^\circ - k_1^\circ + k_2 - k_1\right) \\
&= k_2\left(1 - \frac{k_2^\circ - k_1^\circ}{n}\right) + k_1\left(\frac{k_2^\circ - k_1^\circ}{n}\right) - k_1^\circ + (k_2^\circ - k_1^\circ)\left(1 - \frac{k_2^\circ - k_1^\circ}{n}\right) \\
&\geqslant -k_1^\circ\frac{k_2^\circ - k_1^\circ}{n} + (k_2^\circ - k_1^\circ)\left(1 - \frac{k_2^\circ - k_1^\circ}{n}\right) = (k_2^\circ - k_1^\circ)\left(1 - \frac{k_2^\circ}{n}\right),
\end{aligned}$$

hence there exists $c_2 > 0$ such that

$$\frac{B^{(2)}_{k_1,k_2}}{n} \geqslant c_2 + o(1),$$

proving (5.36) and (5.37).

It is easy to see that analogously to (5.32)

$$\max_{\substack{k_1<k_2 \\ \max(|k_1-k_1^\circ|,|k_2-k_2^\circ|)>N}} \frac{|B^{(2)}_{k_1,k_2}|}{n} = O_P(1),$$

as well as by a case-by-case study as above, for the exemplary case cf. (5.38), analogously to (5.32)

$$\max_{\substack{k_1<k_2 \\ \max(|k_1-k_1^\circ|,|k_2-k_2^\circ|)>N}} \frac{|B^{(1)}_{k_1,k_2}|}{\max(|k_1-k_1^\circ|, |k_2-k_2^\circ|)} = O_P(1).$$

As in the proof of Theorem 3.3 it holds $|\widehat{d}_l - s_l d_l| = o_P(1)$. Analogously to (5.33) we get

$$\max_{\substack{k_1<k_2 \\ \max(|k_1-k_1^\circ|,|k_2-k_2^\circ|)>N}} \frac{|A^{(1)}_{k_1,k_2,l}|}{\max(|k_1-k_1^\circ|, |k_2-k_2^\circ|)} = \alpha(N)^{-1}O_P(1) + o_P(1),$$

as well as analogously to (5.34)

$$\max_{\substack{k_1<k_2 \\ \max(|k_1-k_1^\circ|,|k_2-k_2^\circ|)>N}} \frac{|A^{(2)}_{k_1,k_2,l}|}{n} = o_P(1).$$

The proof can now be completed as the proof of Theorem 3.3. □

5.3. Proofs of Section 4

Proof of Theorem 4.1. For the proof of (a) let γ_l be the eigenvalue of $k(\cdot,\cdot)$ belonging to the w_l and λ_l the eigenvalue of $c(\cdot,\cdot)$ belonging to v_l. We prove the contrapositive. To this end assume

$$\int_{\mathcal{Z}} \Delta(t) w_l(t)\, dt = 0 \quad l = 1, \ldots, d.$$

This implies

$$\begin{aligned}\int c(t,s) w_l(s)\, ds &= \int k(t,s) w_l(s)\, ds - \theta(1-\theta)\Delta(t) \int \Delta(s) w_l(s)\, ds \\ &= \gamma_l w_l(t), \quad l = 1, \ldots, d. \end{aligned} \tag{5.39}$$

This shows that γ_l, $l = 1, \ldots, d$, are eigenvalues of $c(t,s)$ with eigenfunctions w_l. Hence, there exist $r_1, r_2, \ldots, r_d$, $r_s \neq r_t$ for $s \neq t$, such that $\gamma_l = \lambda_{r_l}$ and $w_l = \pm v_{r_l}$.

Recall the min–max principle for the l-largest eigenvalue β_l of a compact non-negative operator Γ in a Hilbert space with inner product $\langle\cdot,\cdot\rangle$ (cf. e.g. [12, Theorem 4.9.1])

$$\beta_l = \min_{\substack{\mathcal{S}\subset\mathcal{Z} \\ \dim(\mathcal{S})=l-1}} \max_{\substack{x\perp\mathcal{S} \\ \|x\|=1}} \langle \Gamma x, x\rangle . \tag{5.40}$$

For the covariance operator it holds

$$\langle Cx, x\rangle = \int_{\mathcal{Z}}\int_{\mathcal{Z}} x(t)c(t,s)x(s)dtds \leqslant \int_{\mathcal{Z}}\int_{\mathcal{Z}} x(t)c(t,s)x(s)dtds + \theta(1-\theta)\left(\int_{\mathcal{Z}} x(t)\Delta(t)\, dt\right)^2 = \langle Kx, x\rangle,$$

where $\langle\cdot,\cdot\rangle$ is the scalar product in the Hilbert space $\mathcal{L}^2(\mathcal{Z})$ and $Cx = \int_{\mathcal{Z}} c(\cdot,s)x(s)\, ds$ and an analogous expression for K. Hence we can conclude from the min–max principle

$$\lambda_l = \min_{\substack{\mathcal{S}\subset\mathcal{Z} \\ \dim(\mathcal{S})=l-1}} \max_{\substack{x\perp\mathcal{S} \\ \|x\|=1}} \langle Cx, x\rangle \leqslant \min_{\substack{\mathcal{S}\subset\mathcal{Z} \\ \dim(\mathcal{S})=l-1}} \max_{\substack{x\perp\mathcal{S} \\ \|x\|=1}} \langle Kx, x\rangle = \gamma_l. \tag{5.41}$$

In particular

$$\lambda_1 \leqslant \gamma_1 = \lambda_{r_1} \leqslant \lambda_1,$$

hence $\lambda_1 = \gamma_1$. Analogously one can deduct inductively that $\lambda_l = \gamma_l, l = 2, \ldots, d$. This implies $w_l = \pm v_l$ and hence $\int \Delta(t) v_l(t)\, dt = \pm \int \Delta(t) w_l(t)\, dt = 0, l = 1, \ldots, d$.

For (b) first note that $k(t, s)/D^2$ has the same eigenfunctions as $k(t, s)$. The eigenvalues are multiples of one another so that the order remains the same and it is sufficient to consider the eigenfunctions of $k(t, s)/D^2$. As $D \to \infty$ we get

$$\frac{k(t,s)}{D^2} = \frac{c(t,s)}{D^2} + \theta(1-\theta)\Delta(t)\Delta(s) \to \theta(1-\theta)\Delta(t)\Delta(s).$$

Since $\Delta(t)\Delta(s)$ has rank 1, it has only one non-zero eigenvalue and the corresponding eigenfunction is $\Delta(\cdot)$. Hence we get by Theorem 2.1

$$\left\| w_{1,D}(\cdot) - s\,\frac{\Delta(\cdot)}{\|\Delta(\cdot)\|} \right\| \to 0.$$

The first assertion immediately follows from this. □

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